

COLBY HILL

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PROFESSIONAL EXPERIENCE

FIDELITY INVESTMENTS

SALT LAKE CITY, UT USA

Senior Full Stack Engineer

October 2023 – Current

Launched an AI powered digital assistant enabling context aware customer conversations at scale, integrating internal systems to drive new capabilities in conversational intelligence and business analytics.

- Architected a patent pending short term memory graph for real time conversation state tracking, reducing computational burden by optimizing information retention, resulting in optimized token usage and enhancing conversation relevance.
- Led the delivery of an analytics platform that parses data from Snowflake, applies LLM analysis to individual interactions, and generates high level insights for strategic decision making and improved customer support.
- Built and integrated a digital assistant with multiple internal systems to engage with customers in context aware conversations through a multi-agent framework and product knowledge graph.

CAPITAL ONE

McLEAN, VA USA

Software Engineer

August 2022 – September 2023

Enhanced customer document processing infrastructure, implementing robust Java microservices and monitoring systems that safeguarded sensitive financial data while enhancing security, preventing fraud, and improving operational efficiency.

- Architected a new Java API to efficiently handle customer uploads and integrate dependent systems, streamlining document flow and improving customer experience.
- Led the modernization of event streaming and logging to Snowflake, Splunk, and New Relic, collaborating with 4 teams cross-functionally, resulting in improved real-time monitoring and analytics.
- Expanded functionality and supported a document platform of Java microservices, facilitating processing of 2 million+ customer document uploads monthly.

CAPITAL ONE

McLean, VA USA

Associate Software Engineer

August 2021 – August 2022

Innovated business credit card rewards redemption experience, implementing new capabilities and improving platform reliability.

- Developed a new Golang API to aggregate customer transactions and determine eligibility for an annual spend bonus, adding a major new capability to promote customer spending.
- Addressed a major customer complaint by implementing a rewards auto-redemption threshold expansion that resulted in a 20% increase in positive customer feedback.
- Led an effort cross functionally to debug and resolve a platform bug which affected customers rewards redemptions for 3 years, improving platform consistency and customer experience.

KRUMWARE

COLUMBIA, SC USA

Software Engineer

August 2019 – May 2020

Full-stack engineer at a tech consulting firm, contributing to 10+ production web apps for clients, a 22% hosting cost reduction via multi-cloud migration, and fully automated Kubernetes resource management.

- Engineered 10+ responsive, user-friendly web applications, consistently exceeding client expectations.
- Led infrastructure migration to Rancher, enabling multi-cloud computing, reducing hosting costs by 22%.
- Automated Kubernetes resource management workflows, eliminating the need for manual processing.

EDUCATION

University of South Carolina

Columbia, SC USA

Master of Science in Computer Science

May 2021

University of South Carolina

Columbia, SC USA

Bachelor of Science in Computer Science

May 2020

TECHNICAL EXPERTISE

- **Languages:** Java, Python, Javascript, Typescript, Neo4J, SQL, NoSQL, C++, Go, Swift, HTML, CSS
- **Technologies:** AWS, Spring, Node.js, Angular, React, Express.js, Django, iOS, Selenium
- **Tools:** Copilot, Claude Code, Docker, Jenkins, New Relic, Splunk, Datadog, Kubernetes, Tableau
- **Certifications:** National Training Standard for Information Systems Security Professionals (January 2020)